

Pulmonary function study
There are no obvious problems with this dataset

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Obs	age	fev	ht	sex	smoke
1	9	1.708	57.0	Female	Nonsmoker
2	8	1.724	67.5	Female	Nonsmoker
3	7	1.720	54.5	Female	Nonsmoker
4	9	1.558	53.0	Male	Nonsmoker
5	9	1.895	57.0	Male	Nonsmoker
6	8	2.336	61.0	Female	Nonsmoker
7	6	1.919	58.0	Female	Nonsmoker
8	6	1.415	56.0	Female	Nonsmoker
9	8	1.987	58.5	Female	Nonsmoker
10	9	1.942	60.0	Female	Nonsmoker

Pulmonary function study

Frequency counts

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The FREQ Procedure

Sex				
sex	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	318	48.62	318	48.62
Male	336	51.38	654	100.00

Smoking status				
smoke	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Nonsmoker	589	90.06	589	90.06
Smoker	65	9.94	654	100.00

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Descriptive statistics

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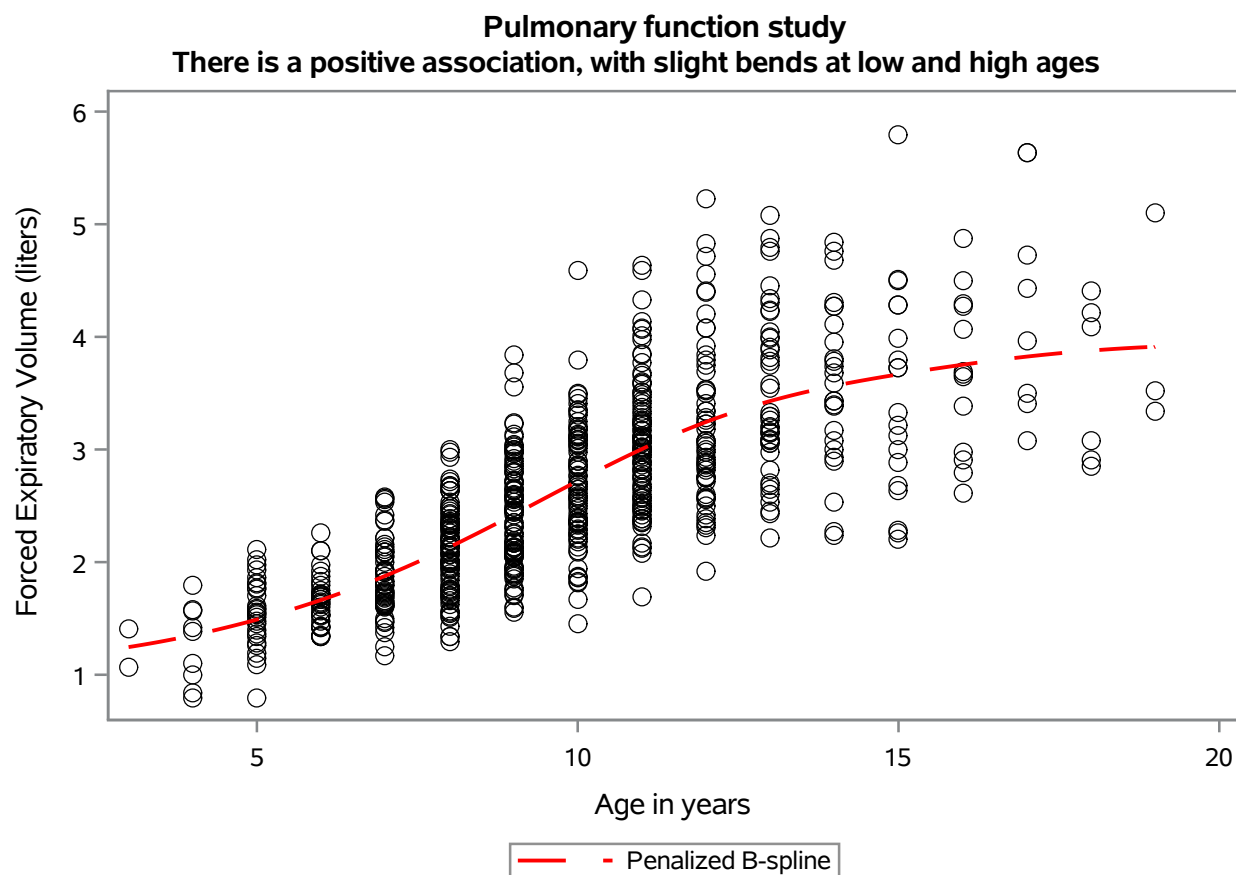
The MEANS Procedure

Variable	Label	N	N Miss	Mean	Std Dev	Minimum	Maximum
age	Age in years	654	0	9.9311927	2.9539352	3.0000000	19.0000000
fev	Forced Expiratory Volume (liters)	654	0	2.6367798	0.8670591	0.7910000	5.7930000
ht	Height in inches	654	0	61.1435780	5.7035128	46.0000000	74.0000000

Pulmonary function study
All variables show a positive correlations

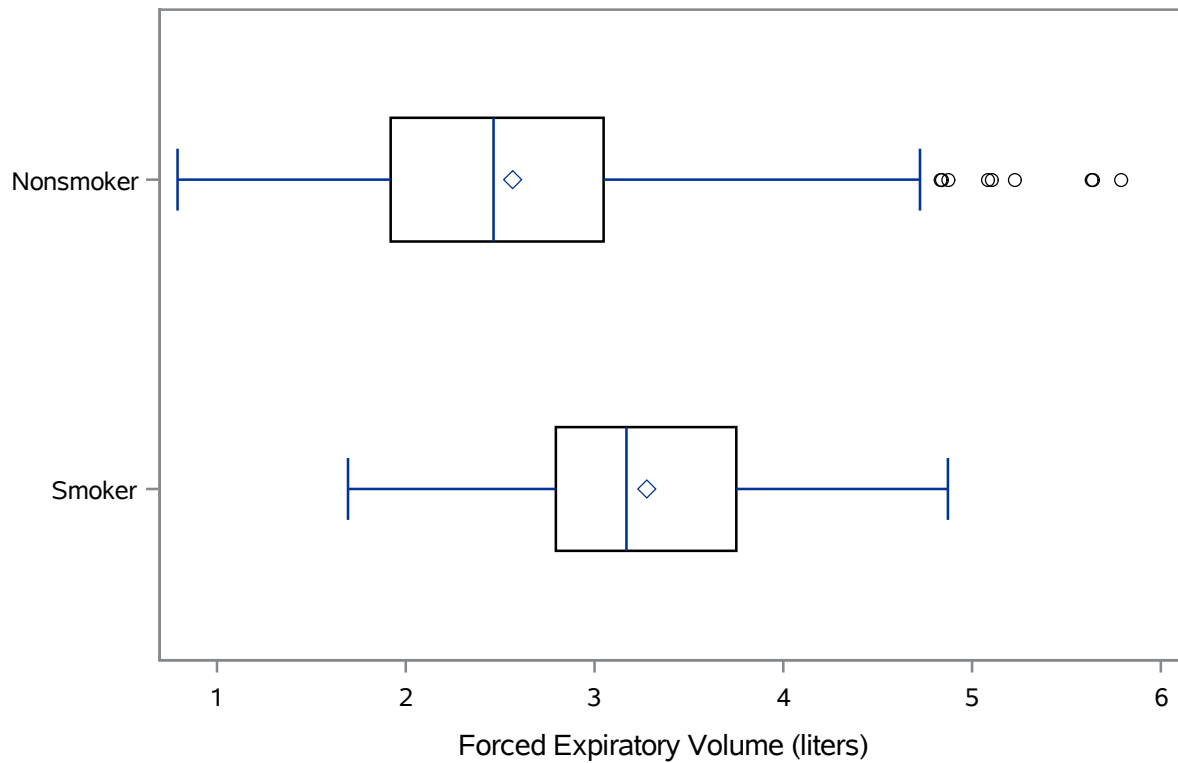
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Obs	_NAME_	age	fev	ht
1	age	1.00	0.76	0.79
2	fev	0.76	1.00	0.87
3	ht	0.79	0.87	1.00

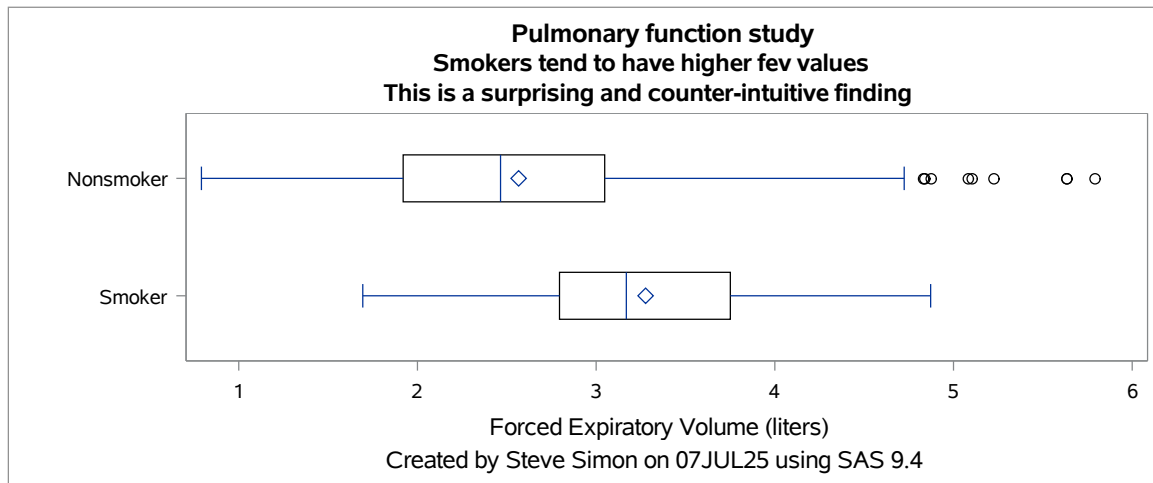


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Pulmonary function study
Smokers tend to have higher fev values
This is a surprising and counter-intuitive finding



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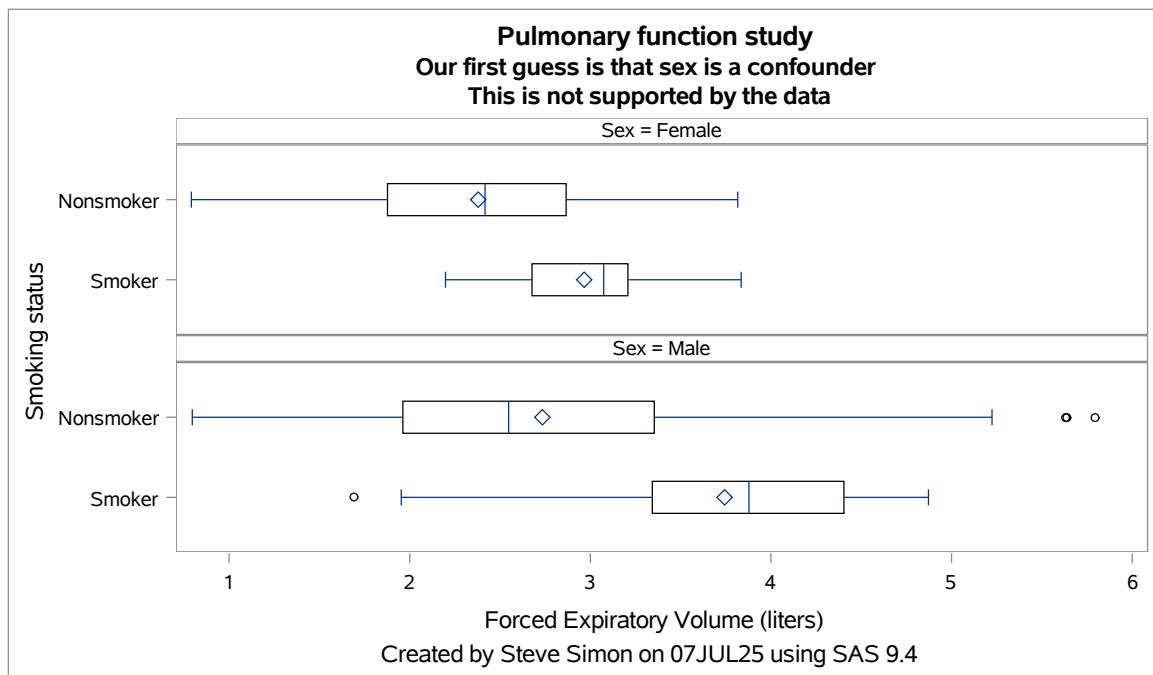
The MEANS Procedure

Smoking status=0

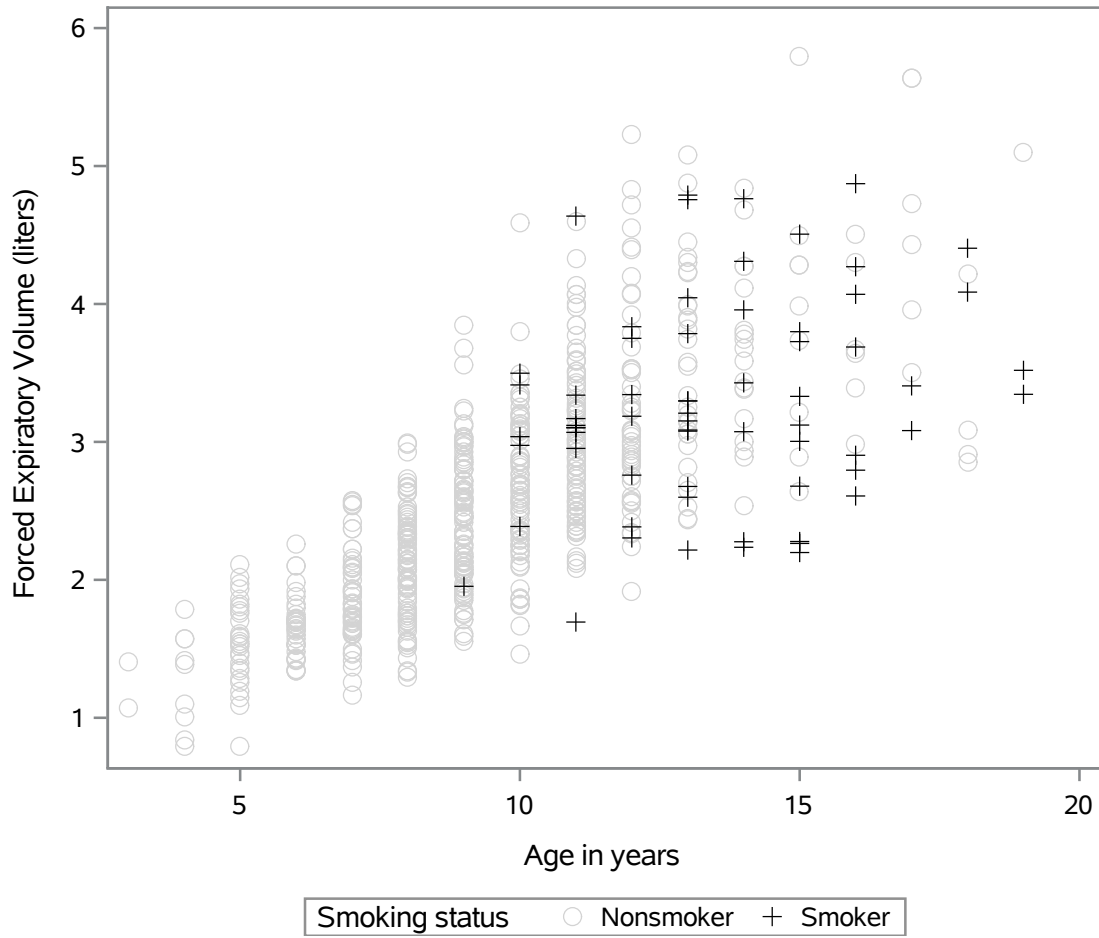
Analysis Variable : fev Forced Expiratory Volume (liters)				
N	Mean	Std Dev	Minimum	Maximum
589	2.5661426	0.8505215	0.7910000	5.7930000

Smoking status=1

Analysis Variable : fev Forced Expiratory Volume (liters)				
N	Mean	Std Dev	Minimum	Maximum
65	3.2768615	0.7499863	1.6940000	4.8720000



Pulmonary function study
Second hypothesis is that age is a confounder
This hypothesis is supported by the data



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